

ESP32 Modular Tool

Multifunctional tool for ESP32 with GC9A01A circular display. Modular menu-driven system with games and utilities.

Features

- **Snake** — classic snake game on a 22x16 grid
- **Stopwatch** — stopwatch with lap times (up to 10)
- **Dice** — configurable dice (d2 to d100)
- **Canvas** — pixel paint with 9 colors
- **Test buttons** — verify each button works
- **Display tests** — fill screen with red/green/blue
- **JPEG viewer** — display JPEG images via Tjpg_Decoder

Hardware

Component	Pin
TFT CS	5
TFT RST	22
TFT DC	21
TFT MOSI	23
TFT SCLK	18
TFT MISO	-1
BTN MENU	16
BTN A	4
BTN B	15
BTN UP	27
BTN DOWN	13
BTN LEFT	14
BTN RIGHT	26
BTN OK	12

Project Structure

```
src/  
├─ main.cpp  
# Setup and loop
```

— ButtonManager/	# Button debounce and callbacks
— ScreenManager/	# All rendering (depends on display libs only)
— GlobalConfig/	# Orchestration, menu, navigation, wiring
— SnakeGame/	# Snake game logic (no rendering)
— CronoManager/	# Stopwatch logic (no rendering)
— DiceManager/	# Dice logic (no rendering)
— CanvasManager/	# Paint logic (no rendering)

Architecture: each **Manager* exposes only state and logic via `const` getters. *ScreenManager* holds all `draw*()` methods. *GlobalConfig* reads manager state and passes it to *ScreenManager* for rendering.

Build

Built with [PlatformIO](#).

```
pio run
pio upload
pio device monitor
```

Dependencies

- Adafruit GFX Library
- Adafruit GC9A01A
- GFX Library for Arduino (TFT_eSPI compatible)
- Tjpg_Decoder